

Article

Aggregation Semantics for Prioritization in Cooperative Automated Decision-Making

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Abstract

Cooperative automated decision-making systems can combine predictive assessments with contextual knowledge to prioritize alternatives under resource constraints. The mechanism used to integrate these sources can influence whether contextual requirements are preserved or can be offset by predictive evidence. This study examines this issue in CADEMAs-ML, a recently proposed framework for cooperative automated decision-making that separates predictive and contextual decision pathways. Three aggregation regimes are analyzed: linear, weighted geometric, and minimum aggregation. The study combines theoretical analysis, controlled Monte Carlo experiments, and an employee-attrition case study. The theoretical analysis characterizes the conditions under which contextual vetoes are preserved during integration. The simulation study evaluates policy compliance, sensitivity to predictive uncertainty, ranking stability, and changes in the Top-K intervention set. The case study applies the same aggregation mechanisms to a reproducible CADEMAs-ML workflow based on predictive models and fuzzy contextual rules. The results show that aggregation semantics can affect policy compliance and prioritization outcomes. Under the considered conditions, zero-absorbing operators preserve contextual vetoes, whereas linear aggregation can allow predictive evidence to compensate for a contextual veto. This effect depends on the relative weighting of the two sources and on the characteristics of the decision population. Predictive uncertainty can also modify rankings and intervention sets, with its effects depending on the aggregation regime and intervention capacity. In the employee-attrition case study, changing only the aggregation mechanism produces different intervention sets from the same predictive and contextual assessments. These findings support treating aggregation as an explicit design choice in cooperative automated decision-making, since its semantics influence how contextual requirements are translated into automated prioritization.



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1. Introduction

Automated decision-making increasingly relies on the integration of heterogeneous sources of evidence [1,2]. Predictive models can estimate the likelihood or severity of an outcome, while contextual knowledge can express organizational requirements, constraints,

preferences, or policy considerations that are not necessarily captured by the predictive model itself [3–5]. This combination is particularly relevant in context-aware automated decision-making, where the decision process is expected to account for both data-driven assessments and the context in which a decision is made. Cooperative automated decision-making extends this idea by allowing partially independent decision-making components to contribute different sources of evidence to a common decision process [6]. Such architectures can support applications in which a limited number of alternatives must be prioritized for intervention, making the integration of the available evidence an important part of the decision policy.

One example of this approach is CADEMAs-ML [7], a context-aware cooperative automated decision-making framework [6] that combines predictive and contextual assessments before producing a final prioritization. In this type of system, predictive components and contextual components may be developed independently and may represent different aspects of the decision problem. Their outputs must then be integrated into a common score or ranking. This final integration step is usually done with an aggregation function, but its consequences can be more substantive. Different aggregation functions can encode different assumptions about how predictive evidence and contextual requirements should interact. Consequently, the aggregation mechanism can affect not only the numerical value of the final score, but also the alternatives that are prioritized when the available intervention capacity is limited.

This raises a central question for cooperative automated decision-making: How does the choice of aggregation semantics affect the resulting prioritization when predictive and contextual evidence are combined? The question is especially relevant when contextual information represents requirements that should constrain the final decision rather than simply contribute another compensatory source of evidence. An aggregation rule that allows one source to compensate for another may produce a different decision from a rule that gives precedence to contextual restrictions. In a resource-constrained scenario, these differences can determine which alternatives receive an intervention. The aggregation mechanism, therefore, deserves consideration as part of the decision policy itself rather than only as a technical step performed after the predictive and contextual assessments have been obtained [8–11].

The issue is further complicated when uncertainty or noise in predictive assessments exists. Prediction errors can change the relative ordering of alternatives and may, therefore, propagate from the score level to the actual set selected for intervention. The magnitude and nature of this effect can depend on the aggregation mechanism, its parameters, and the intervention capacity. A meaningful assessment of aggregation choices should, therefore, consider both their structural properties and their observable consequences for policy compliance, ranking stability, and resource-constrained Top-K selection [12,13].

Research on aggregation functions and fuzzy multi-criteria decision-making (MCDM) provides a broad range of mechanisms for combining heterogeneous criteria and expressing compensatory and non-compensatory decision behavior [8–11,14]. Likewise, research on cooperative and context-aware automated decision-making provides architectures for combining multiple sources of decision evidence [3,4,6,15]. The connection between these two perspectives raises a more specific issue that has received less attention in the CADEMAs-ML setting: how the structural properties of an aggregation operator translate into concrete policy and prioritization outcomes when predictive and contextual assessments are integrated. In particular, the effects on contextual policy compliance, sensitivity to predictive uncertainty, and the composition of alternatives selected under a fixed intervention budget require explicit analysis.

This paper addresses this issue by analyzing the integration of predictive and contextual scores in CADEMAs-ML through three representative aggregation regimes: a compensatory linear operator, a weighted geometric operator, and a minimum operator. The analysis combines formal results with controlled Monte Carlo experiments and a reproducible employee-attrition case study. The theoretical analysis establishes conditions under which contextual requirements are preserved or can be overridden during integration. The simulation study examines policy compliance, the effect of predictive-score perturbations, ranking agreement, and the dependence of observed violations on the Top-K intervention capacity. Finally, the case study applies the same aggregation alternatives to a CADEMAs-ML workflow in which predictive and contextual assessments are generated independently before being integrated into the final prioritization [6,7].

The main contributions can be summarized as follows:

- A formal characterization of the relationship between aggregation semantics and contextual veto preservation, identifying zero absorption as the key structural property for maintaining a zero contextual score through the integration stage.
- A theoretical and empirical analysis of how compensatory and zero-absorbing aggregation operators affect policy compliance and resource-constrained prioritization under controlled population conditions.
- A sensitivity analysis of prioritization under predictive uncertainty, including its effects on policy violations, ranking stability, inter-operator agreement, and the composition of Top-K intervention sets.
- A reproducible employee-attrition case study illustrating the practical consequences of aggregation semantics and showing how changing only the integration mechanism can alter the composition of a fixed intervention budget.

The remainder of the paper is organized as follows. Section 2 presents the background and related work, covering cooperative automated decision-making, fuzzy MCDM, and aggregation theory, and formulates the research questions addressed in this study. Section 2.2 then introduces the CADEMAs architecture and its CADEMAs-ML specialization, providing the context for the aggregation problem examined here. Section 4 presents the considered aggregation operators and analyzes their structural properties and policy implications. The resulting effects are first examined under controlled conditions in Section 5, and subsequently illustrated in an employee-attrition case study in Section 6. Finally, Section 7 is devoted to discussion and conclusions, while also outlines directions for future research.

2. Background and Related Work

This section establishes the conceptual and methodological foundations of the study. It first introduces the cooperative automated decision-making setting in which predictive and contextual assessments are integrated. Secondly, the Cooperative Automated Decision-Making System (CADEMAs) framework and its specialization for machine-learning-based units are both described. Thirdly, the section presents some aspects about fuzzy MCDM and aggregation theory. Particular attention is given to aggregation properties that determine whether evidence from one decision pathway can compensate for deficiencies in another.

2.1. Context-Aware and Cooperative Decision-Support Systems

The increasing adoption of Artificial Intelligence (AI) in decision-support systems (DSS) has highlighted the need to complement predictive performance with contextual, organizational, and domain-specific knowledge. In many real-world settings, decisions cannot be based solely on statistical predictions because institutional policies, expert knowledge, regulatory requirements, and operational constraints must also be consid-

ered. As a result, a growing body of research has focused on context-aware and cooperative decision-support systems capable of integrating heterogeneous sources of information into the decision-making process.

One important research direction concerns the incorporation of explicit knowledge representations alongside data-driven models. Early examples include cognitive-agent architectures that combine symbolic rules with machine-learning mechanisms to support context-aware decision-making [3]. Similarly, ontology-based clinical decision-support systems have employed contextual ontologies and clinical pathways to select context-appropriate recommendations for individual cases [4]. More recent developments have extended these ideas through semantic models, knowledge graphs, and hybrid AI architectures that combine predictive models with structured domain knowledge to improve decision quality and interpretability [16,17].

A second research stream emphasizes cooperation between human and artificial intelligence. Rather than viewing decision support as a purely predictive task, these approaches consider decision-making as a collaborative process in which machine-learning models provide evidence while human experts contribute contextual understanding, oversight, and governance. Cronholm and Göbel [1], for example, argue that effective decision support requires appropriate mechanisms for combining human and artificial intelligence rather than simply improving predictive accuracy. Likewise, recent surveys on human-centered AI highlight transparency, accountability, contextual awareness, and governance as fundamental requirements for trustworthy decision-support systems [18].

Related ideas also appear in cooperative and distributed decision-making environments. Multi-agent systems, collective-intelligence frameworks, and cooperative AI architectures frequently rely on multiple decision sources that must be reconciled before a final recommendation is produced [15]. In these settings, contextual information often plays a coordinating role by constraining, guiding, or validating the outputs generated by predictive components. Similar concerns arise in governance-oriented decision-support systems, where contextual information may encode operational procedures, organizational priorities, safety requirements, or regulatory obligations [17,19].

Taken together, these studies demonstrate the importance of contextual knowledge in decision-support systems. However, their primary focus typically lies in knowledge representation, ontology design, human–AI interaction, governance mechanisms, or system architecture. Once predictive evidence and contextual assessments have been generated, the numerical integration process that combines both sources of information is often treated as a secondary implementation detail. Consequently, relatively little attention has been devoted to understanding how alternative integration mechanisms affect the final recommendations produced by the system.

2.2. CADEMAs and CADEMAs-ML

The Cooperative Automated Decision-Making System (CADEMAS) framework was introduced by Novoa-Hernández et al. [6] as a general architecture for context-aware decision support in which multiple heterogeneous ADM units cooperate to produce a unified recommendation, score, or decision. Rather than prescribing a specific decision model, CADEMAs provides a modular structure that separates the generation of decision evidence from its contextual evaluation and final integration.

Many decision-support systems combine predictive evidence with contextual information, but these elements are often intertwined within the same model or decision process. In contrast, CADEMAs explicitly distinguishes three conceptually different functions: (i) evidence generation, (ii) contextual assessment, and (iii) decision integration. Conse-

quently, the aggregation mechanism can be analyzed independently of both the predictive subsystem and the contextual-knowledge representation.

Formally, a CADEMAs instance is represented by the tuple

$$S = \langle I, U, O, C, T \rangle, \quad (1)$$

where I denotes the Input Manager, U the set of ADM units, O the Decision Integration Layer, C the Contextual Modulator, and T the universe of admissible data representations.

Given an input case, the ADM units independently generate decision evidence from the available information. In parallel, the Contextual Modulator evaluates the extent to which the same case satisfies organizational priorities, expert knowledge, regulatory requirements, or other contextual constraints (i.e., the “context”). The outputs of both subsystems are then combined by the Decision Integration Layer to produce the final decision score.

An important specialization of this framework is CADEMAs-ML [7], in which the ADM units correspond to machine-learning models. In this setting, multiple predictive models cooperate to produce a consolidated predictive assessment, while contextual information is represented through a fuzzy inference system inspired by Pérez-Cañedo et al. [5] and Pérez-Cañedo et al. [20]. The contextual subsystem translates organizational priorities and institutional requirements into a normalized degree of contextual satisfaction.

To focus on the role of aggregation, the internal implementation of both subsystems is abstracted in terms of two normalized scores. For each decision case or alternative i , the predictive subsystem produces a predictive score

$$R_i \in [0, 1], \quad (2)$$

which summarizes the evidence generated by the predictive ADM units. Depending on the application domain, R_i may represent a risk estimate, a confidence score, a probability of occurrence, or any other normalized predictive assessment.

Independently, the contextual subsystem produces a context score

$$Q_i \in [0, 1], \quad (3)$$

which quantifies the degree to which the case satisfies the contextual requirements.

It is worth noting that the restriction of both predictive and contextual scores to the unit interval does not limit the generality of the proposed formulation.

Consequently, R_i and Q_i should be interpreted as normalized semantic scores rather than measurements expressed in a physical unit. The use of the unit interval allows the analysis of aggregation properties independently of the original scale of the predictive and contextual information. More importantly, the lower boundary $Q_i = 0$ represents a complete contextual violation.

A final prioritization score is then obtained through an aggregation operator

$$P_i = A(R_i, Q_i), \quad (4)$$

where A defines the integration semantics between predictive evidence and contextual assessment. Candidate cases are subsequently ranked according to P_i , and the corresponding Top-K subset is selected for intervention or resource allocation.

It is important to emphasize that the aggregation problem considered here differs from traditional fuzzy MCDM settings. In classical MCDM, aggregation operators combine multiple criteria describing different attributes of the same alternative, typically with the objective of capturing trade-offs among decision criteria. In CADEMAs-ML, however, the operator

combines heterogeneous sources of decision evidence that play fundamentally different roles: a predictive assessment generated by cooperating machine-learning models and a contextual assessment derived from organizational, regulatory, or domain-specific knowledge.

Consequently, different aggregation semantics can be introduced without modifying either the predictive subsystem or the contextual subsystem. This modularity provides an ideal setting for studying aggregation as a policy-design mechanism and for analyzing how different compensation regimes affect prioritization, policy compliance, robustness, and ranking stability.

2.3. Aggregation Operators and Decision Semantics

Aggregation operators constitute one of the central building blocks of multi-criteria decision-making algorithms (MCDM), fuzzy decision support, and collective decision-making systems. Their purpose is to combine multiple sources of information into a single assessment while preserving, to a greater or lesser extent, the decision semantics associated with the original criteria.

The literature about aggregation operators is huge and reviewing it is outside of the scope of this work. Among the most influential aggregation frameworks are Ordered Weighted Averaging (OWA) operators [8,9], but many other are available, like t-norm- and t-conorm-based operators, geometric aggregation functions, and non-additive integrals such as the Choquet and Sugeno integrals. The interested reader may refer to [21,22].

More recently, reviews by Mardani et al. [10] and Csiszár [11] highlight the widespread use of OWA-based aggregation and related fuzzy operators across diverse application domains. More recent studies have continued to investigate how aggregation semantics influence consensus formation, robustness, criterion interaction, and ranking behavior. Examples include consensus-oriented aggregation in group decision-making [23,24], robust Choquet-based decision models [12,13], and applications employing geometric and OWA-type operators to capture alternative decision attitudes [14].

Most of these studies are formulated within classical MCDM settings, where the aggregated inputs represent multiple criteria, attributes, preferences, or expert judgments associated with the same decision problem. The primary objective is, typically, to identify the most appropriate operator for combining these criteria or to develop new aggregation mechanisms with desirable mathematical properties. Comparatively less attention has been paid to architectures in which the aggregated inputs originate from fundamentally different decision layers.

This distinction is particularly important in cooperative automated decision-making systems. In architectures such as CADEMAs-ML, the aggregation process does not combine multiple criteria describing an alternative. Instead, it reconciles two conceptually distinct sources of evidence: a predictive assessment generated by machine-learning models and a contextual assessment encoding organizational priorities, regulatory requirements, expert knowledge, or policy constraints. Under this interpretation, the aggregation operator effectively determines the extent to which predictive evidence may compensate for weak contextual support or, conversely, whether contextual constraints can restrict otherwise favorable predictions. The operator, therefore, acts as an implicit governance mechanism rather than merely a criterion-combination tool.

For this reason, the present work focuses on three representative aggregation regimes exhibiting fundamentally different compensation semantics. The weighted arithmetic mean represents a fully compensatory integration strategy in which predictive and contextual information may offset one another. The weighted geometric mean represents a partially compensatory strategy that penalizes strong imbalances between both sources of evidence. Finally, the minimum operator represents a non-compensatory strategy in which the

weakest component determines the final assessment. Together, these operators span a spectrum ranging from highly permissive to highly restrictive integration behavior while remaining interpretable and widely understood within the aggregation literature.

3. Research Gap, Positioning and Research Questions

The previous section helped to identify two largely independent research streams. On the one hand, context-aware and cooperative decision-support systems have developed increasingly sophisticated mechanisms for incorporating contextual knowledge into predictive processes through ontologies, symbolic rules, knowledge graphs, human oversight, and governance-oriented architectures. These studies demonstrate the importance of contextual information but typically focus on knowledge representation, rule selection, system architecture, or human–AI interaction. Once predictive and contextual assessments have been generated, the mechanism responsible for their numerical integration is often treated as an implementation detail.

On the other hand, the literature on aggregation theory and fuzzy MCDM provides extensive theoretical knowledge regarding compensation, criterion interaction, robustness, consensus formation, and preference aggregation. However, aggregation operators are generally studied as mathematical constructs or as components of multi-criteria decision methods, where the aggregated inputs represent criteria, preferences, or expert judgments associated with the same decision problem.

Consequently, a gap exists between context-aware decision-support frameworks and aggregation theory. The former provides mechanisms for representing contextual knowledge but rarely investigates how different aggregation semantics affect the preservation of contextual constraints after integration. The latter offers powerful aggregation mechanisms but seldom studies their implications when heterogeneous decision signals, such as predictive machine-learning outputs and contextual assessments, must jointly determine operational recommendations.

From this perspective, the gap addressed in this work is not the absence of aggregation operators, nor the lack of mathematical knowledge about their properties. Rather, it is the lack of a systematic understanding of aggregation as a governance mechanism within cooperative automated decision-making architectures. In particular, relatively little attention has been devoted to questions such as whether contextual assessments can effectively act as veto mechanisms, how different compensation regimes influence policy compliance, how sensitive prioritization outcomes are to uncertain predictive scores, or how ranking stability is affected by uncertainty in contextual information.

These observations motivate the following research questions:

- *RQ1. How do different aggregation semantics affect the interaction between predictive evidence and contextual assessments in a cooperative automated decision-making architecture?*
- *RQ2. Under what conditions can aggregation operators preserve or violate contextual constraints and veto-like policies?*
- *RQ3. How do alternative aggregation regimes influence contextual policy compliance and prioritization robustness under uncertain predictive scores?*
- *RQ4. To what extent do different aggregation semantics alter the stability of the resulting rankings and intervention priorities?*

4. Aggregation Semantics and Theoretical Analysis

This section provides the theoretical foundations required to answer the research questions introduced in Section 3, particularly those related to the role of aggregation semantics in cooperative automated decision-making. Specifically, the analysis addresses how different aggregation regimes modify the interaction between predictive and con-

textual information (RQ1), under which conditions contextual constraints and veto-like policies are preserved or violated (RQ2), how aggregation semantics influence contextual policy compliance and robustness under uncertain predictive scores (RQ3), and how they affect the stability of the resulting prioritization rankings (RQ4).

The section is organized in three parts. It first formalizes the structural properties that characterize aggregation behavior, including boundedness, monotonicity, continuity, zero absorption, contextual veto preservation, and compensability. It then examines three representative aggregation regimes with different decision semantics: fully compensatory linear aggregation, partially compensatory aggregation through the weighted geometric mean, and non-compensatory minimum aggregation. It closes by deriving conditions for contextual veto preservation, rank reversal under compensation, and sensitivity to perturbations in predictive and contextual assessments.

4.1. Structural Properties of Aggregation Operators

Without loss of generality, all scores are assumed to be normalized within the unit interval. Let

$$A : [0, 1]^2 \rightarrow [0, 1] \quad (5)$$

denote an aggregation operator that combines the predictive score R_i and the contextual score Q_i into the final prioritization score

$$P_i = A(R_i, Q_i). \quad (6)$$

Definition 1 (Boundedness). *An aggregation operator is bounded if*

$$0 \leq A(r, q) \leq 1, \quad \forall (r, q) \in [0, 1]^2. \quad (7)$$

Boundedness guarantees that the resulting prioritization score remains within the same normalized scale as the predictive and contextual assessments. Since A maps $[0, 1]^2$ into $[0, 1]$, this property is automatically satisfied and is stated here only for completeness.

Definition 2 (Monotonicity). *An aggregation operator is monotonic if*

$$r_1 \leq r_2, \quad q_1 \leq q_2 \implies A(r_1, q_1) \leq A(r_2, q_2). \quad (8)$$

Monotonicity ensures that improving either predictive evidence or contextual satisfaction cannot decrease the final prioritization score.

Definition 3 (Continuity). *An aggregation operator is continuous if small perturbations in its inputs produce arbitrarily small changes in the aggregated score.*

Continuity is particularly relevant when predictive and contextual scores are estimated from noisy data, since it prevents abrupt changes in prioritization due to infinitesimal variations in the inputs.

Definition 4 (Zero Absorption). *An aggregation operator satisfies the zero-absorption property if*

$$A(r, 0) = 0, \quad \forall r \in [0, 1]. \quad (9)$$

Under zero absorption, a null contextual assessment completely suppresses the final prioritization score regardless of the predictive evidence available.

Definition 5 (Strict Positivity on the Interior). *An aggregation operator is strictly positive on the interior if*

$$A(r, q) > 0, \quad \forall (r, q) \in (0, 1)^2. \quad (10)$$

Zero absorption and strict positivity play complementary roles. The first forces a null score exactly when the context vetoes an alternative. The second guarantees that no other alternative is driven to zero. Taken together, they make the value $P_i = 0$ an unambiguous marker of a contextual veto, which is what allows veto preservation to be read directly from the aggregated scores.

Definition 6 (Contextual Veto Preservation). *An aggregation operator A preserves contextual vetoes if, for every decision case i ,*

$$Q_i = 0 \implies P_i = A(R_i, Q_i) = 0.$$

Within the CADEMAs-ML setting, contextual veto preservation means that contextual requirements cannot be overridden by predictive evidence. Because R_i may take any value in $[0, 1]$, contextual veto preservation is equivalent to zero absorption (Definition 4).

Definition 7 (Compensability). *In this work, an aggregation operator is compensatory on an interior region if a decrease in one input can be locally offset by an increase in the other while preserving the aggregated score. For a differentiable operator, a sufficient local characterization is that both partial derivatives are strictly positive in that region.*

Thus, compensability denotes local substitutability rather than the ability to offset an arbitrary deficit. The unit-scale bounds limit every trade-off. It determines the extent to which strong predictive evidence may compensate for weak contextual support, or vice versa. At one extreme, fully compensatory operators permit the broadest within-scale trade-offs between prediction and context. At the opposite extreme, non-compensatory operators such as the minimum prevent such trade-offs; when combined with zero absorption, this behavior implements a hard contextual veto.

These properties provide a theoretical basis for distinguishing alternative integration semantics within CADEMAs-ML. In particular, zero absorption and compensability play a central role in determining whether contextual requirements behave as soft preferences, partial constraints, or strict vetoes. The following subsection introduces the specific aggregation operators considered in this work and analyzes how they instantiate these structural properties.

4.2. Aggregation Operators Considered

To investigate how aggregation semantics affect prioritization behavior, policy compliance, and robustness, this work considers three representative operators that span different levels of compensability between predictive evidence and contextual requirements.

1. *Linear aggregation.* The linear operator corresponds to the integration mechanism adopted in existing CADEMAs-ML implementations [7]:

$$A_L(R_i, Q_i) = \lambda R_i + (1 - \lambda) Q_i, \quad \lambda \in [0, 1]. \quad (11)$$

The parameter λ controls the relative importance assigned to predictive and contextual information. For $\lambda \in (0, 1)$, this operator is compensatory: reductions in one component can be offset by increases in the other, subject to the bounds of the unit

interval. The boundary cases $\lambda = 0$ and $\lambda = 1$ are degenerate and are not considered as compensatory regimes in the subsequent analysis.

2. *Weighted geometric aggregation.* An intermediate alternative is the weighted geometric mean:

$$A_G(R_i, Q_i) = R_i^\lambda Q_i^{1-\lambda}, \quad \lambda \in (0, 1). \quad (12)$$

Unlike the linear operator, the weighted geometric mean penalizes imbalanced evaluations and, therefore, reduces the degree of compensation between predictive and contextual information. Nevertheless, partial compensation remains possible whenever both inputs are strictly positive. Restricting the exponent to $\lambda \in (0, 1)$ keeps both powers strictly between 0 and 1, so the indeterminate form 0^0 never arises and $A_G(r, 0) = 0$ holds for every r . The two excluded endpoints behave differently: at $\lambda = 1$ the operator reduces to $A_G(R_i, Q_i) = R_i$ and loses the veto property, and at $\lambda = 0$ it reduces to Q_i . Both are, therefore, excluded from the analysis, and the implementation assigns $A_G = 0$ whenever either input equals zero.

3. *Minimum aggregation.* The third regime is the minimum operator

$$A_M(R_i, Q_i) = \min(R_i, Q_i). \quad (13)$$

This operator represents a non-compensatory integration regime in which the final prioritization score cannot exceed the weakest component. The minimum operator is also a t-norm and, therefore, provides a canonical conjunctive aggregation mechanism. Consequently, predictive evidence cannot compensate for insufficient contextual support. Note that A_M carries no weighting parameter. Its output is invariant under changes in λ , so any comparison across integration regimes leaves the minimum operator unchanged by construction. This point matters when reading the λ -dependent results in Sections 5.2 and 5.3.

Table 1 summarizes the structural properties of the three operators according to the definitions introduced in the previous subsection.

Table 1. Structural properties of the aggregation operators considered in this work.

Property	Linear	Geometric	Minimum
Boundedness	Yes	Yes	Yes
Monotonicity	Yes	Yes	Yes
Continuity	Yes	Yes	Yes
Zero absorption	No [†]	Yes	Yes
Contextual veto preservation	No [†]	Yes	Yes
Strict positivity on $(0, 1]^2$	Yes	Yes	Yes
Compensability	Full [‡]	Partial	No

[†] For the linear operator, zero absorption and contextual veto preservation hold only in the degenerate case $\lambda = 0$. [‡] Full compensability for the linear operator is understood for $\lambda \in (0, 1)$, with compensation bounded by the $[0, 1]$ scale.

For non-degenerate weights $\lambda \in (0, 1)$, the three operators are strictly positive on the interior and differ in terms of zero absorption. This isolates the property responsible for the contrasting policy behavior reported in Section 5: the operators agree on which alternatives deserve a positive score and disagree only on whether a contextual veto can be overridden.

These operators were selected because they represent three distinct integration semantics: fully compensatory aggregation through the linear operator, partially compensatory non-linear aggregation through the weighted geometric operator, and non-compensatory conjunctive aggregation through the minimum t-norm. The selection is, therefore, semantic

rather than exhaustive: together, the three operators cover distinct positions along the compensation–constraint spectrum while allowing the role of zero absorption to be isolated. The linear and weighted geometric operators provide compensatory alternatives with different responses to imbalanced assessments, whereas the minimum operator represents a strict contextual constraint.

This semantic scope also motivates the exclusion of other operators like disjunctive aggregation. In particular, the maximum operator, $A_{\max}(R, Q) = \max(R, Q)$, represents an optimistic integration regime in which the stronger input determines the prioritization score. As a result, high predictive evidence can compensate for weak contextual support, and a contextual veto ($Q_i = 0$) does not suppress the aggregate whenever $R_i > 0$. Such behavior is inconsistent with the policy-preserving role of contextual information considered in CADEMAs-ML, where contextual requirements may constrain or veto prioritization. The analysis, therefore, focuses on aggregation regimes that allow compensation to varying degrees while remaining compatible with this constraint-oriented setting.

The following subsections analyze the theoretical consequences of these semantics and their implications for prioritization behavior. Here, *full within-scale compensation* refers to the linear operator with $\lambda \in (0, 1)$; it does not mean unbounded compensation.

4.3. Conditions for Contextual Veto Preservation

In some applications, a contextual assessment should not behave as a soft preference but rather as a hard constraint. This situation occurs when a complete contextual violation must prevent an alternative from being prioritized, regardless of its predictive evidence.

By Definitions 4 and 6, an aggregation operator preserves contextual vetoes exactly when it satisfies the zero-absorption condition (9). Therefore, the question of whether CADEMAs-ML can enforce contextual vetoes reduces to one of determining whether the selected aggregation operator satisfies zero absorption.

Theorem 1 (Contextual veto preservation). *Let $A : [0, 1]^2 \rightarrow [0, 1]$ be an aggregation operator used in CADEMAs-ML. If A satisfies zero absorption, then every decision case i with a null contextual assessment,*

$$Q_i = 0, \quad (14)$$

receives a null prioritization score,

$$P_i = A(R_i, Q_i) = 0, \quad (15)$$

independently of its predictive score R_i .

Proof. The result follows directly from the definition of zero absorption. Since $Q_i = 0$, it follows that $P_i = A(R_i, Q_i) = A(R_i, 0)$. Because A satisfies zero absorption, $A(R_i, 0) = 0$, for every $R_i \in [0, 1]$. Therefore, $P_i = 0$. Hence, a complete contextual violation suppresses the final prioritization score regardless of the predictive evidence. $\square$

Corollary 1 (Veto preservation of the considered operators). *Among the aggregation operators considered in this work, the minimum and weighted geometric operators preserve contextual vetoes, whereas the linear operator does not for $\lambda \in (0, 1]$, with the degenerate case $\lambda = 0$ being an exception.*

Proof. For the minimum operator, $A_M(r, 0) = \min(r, 0) = 0$. Similarly, for the weighted geometric operator, $A_G(r, 0) = r^\lambda 0^{1-\lambda} = 0$, for $\lambda \in (0, 1)$. In contrast, for $\lambda \in (0, 1]$, the linear operator yields $A_L(r, 0) = \lambda r$, which is greater than zero whenever $r > 0$ and $\lambda > 0$. Therefore, predictive evidence can override a complete contextual violation under

linear aggregation. For the degenerate case $\lambda = 0$, $A_L(r, 0) = 0$, so the linear operator trivially preserves contextual vetoes. $\square$

Theorem 1 constrains individual prioritization scores. In a Top-K selection, however, the relevant question is whether these score-level guarantees are sufficient to exclude vetoed alternatives from the selected set. The following corollary makes this connection explicit and identifies the additional population-level condition required.

Corollary 2 (Exclusion of vetoed alternatives from the Top-K set). *Let*

$$\mathcal{V} = \{i : Q_i = 0\}$$

denote the set of vetoed alternatives, and let $\mathcal{T}_K$ denote the set of alternatives selected among the K largest prioritization scores, for a fixed Top-K capacity K . Let A satisfy zero absorption and strict positivity on the interior (Definition 5). Consider a population of N alternatives, of which N_V are vetoed ($Q_i = 0$), while every remaining alternative satisfies $R_i > 0$ and $Q_i > 0$. If

$$N - N_V \geq K,$$

then

$$\mathcal{T}_K \cap \mathcal{V} = \emptyset.$$

Proof. By zero absorption, every vetoed alternative receives a zero prioritization score,

$$P_i = A(R_i, 0) = 0, \quad i \in \mathcal{V}.$$

By strict positivity on the interior, every non-vetoed alternative satisfying $R_i > 0$ and $Q_i > 0$ receives a strictly positive score,

$$P_i = A(R_i, Q_i) > 0, \quad i \notin \mathcal{V}.$$

Since $N - N_V \geq K$, there are at least K non-vetoed alternatives with strictly positive prioritization scores. Consequently, the K largest scores are all strictly positive, whereas every vetoed alternative has a score of zero. Hence, no vetoed alternative can belong to the Top-K set, and, therefore,

$$\mathcal{T}_K \cap \mathcal{V} = \emptyset.$$

$\square$

The corollary shows that zero absorption provides a score-level guarantee that extends to Top-K selection when a sufficient number of admissible alternatives is available. Thus, zero absorption alone is not sufficient to characterize the selected set: a population-level supply condition, $N - N_V \geq K$, is also required. If fewer than K non-vetoed alternatives are available, some vetoed alternatives may necessarily remain in the selected set, even though the aggregation operator assigns them a zero prioritization score.

This observation motivates the subsequent analysis of rank-reversal conditions and compensation thresholds, which characterize how much contextual improvement is required to offset predictive advantages under different aggregation regimes.

4.4. Conditions for Rank Reversal Under Compensation

The previous section established that contextual veto preservation depends on the existence of a zero-absorption property. However, many aggregation operators follow a different philosophy: they allow partial compensation between the aggregated sources of

evidence. In such cases, a sufficiently high value in one dimension may offset a lower value in another dimension.

This property is common in compensatory aggregation schemes, including weighted arithmetic means and several non-linear aggregation operators. Although compensation provides flexibility when predictive evidence and contextual information represent partially substitutable criteria, it also introduces the possibility that the final prioritization ranking differs from the ranking induced by predictive evidence alone.

To formalize this phenomenon, let $A : [0, 1]^2 \rightarrow [0, 1]$ be a continuous aggregation operator that is strictly increasing in both arguments. Given two alternatives i and j , assume that alternative i has stronger predictive evidence, but alternative j has better contextual alignment: $R_i > R_j$ and $Q_i < Q_j$. The final prioritization scores are, then, $P_i = A(R_i, Q_i)$ and $P_j = A(R_j, Q_j)$. A rank reversal occurs when $P_j > P_i$, meaning that the contextual advantage of alternative j compensates for its predictive disadvantage.

Theorem 2 (General rank-reversal condition under compensation). *Let $A : [0, 1]^2 \rightarrow [0, 1]$ be a continuous aggregation operator strictly increasing in both arguments. For two alternatives i and j satisfying $R_i > R_j$ and $Q_i < Q_j$, define $\Phi_A(R_i, R_j, Q_i)$ as the unique solution to*

$$A(R_j, Q_i + \Phi_A) = A(R_i, Q_i), \quad (16)$$

when such a solution exists in $(0, 1 - Q_i]$. Then, a rank reversal in favor of alternative j occurs if and only if

$$Q_j > Q_i + \Phi_A(R_i, R_j, Q_i). \quad (17)$$

Proof. Fix $R_i > R_j$ and Q_i . Define the function $g(q) = A(R_j, q) - A(R_i, Q_i)$. Because A is continuous and strictly increasing in its second argument, $g(q)$ is also continuous and strictly increasing. At $q = Q_i$, $g(Q_i) = A(R_j, Q_i) - A(R_i, Q_i) < 0$ because A is strictly increasing in its first argument and $R_j < R_i$. If the condition of the theorem is satisfied, there exists a unique $\Phi_A \in (0, 1 - Q_i]$ such that $g(Q_i + \Phi_A) = 0$, which is equivalent to $A(R_j, Q_i + \Phi_A) = A(R_i, Q_i)$. Therefore, $g(Q_j) > 0$ (i.e., $A(R_j, Q_j) > A(R_i, Q_i)$) holds if and only if $Q_j > Q_i + \Phi_A$, since g is strictly increasing. This proves the claimed equivalence. $\square$

Under its stated feasibility condition, the theorem shows that rank reversal need not be an artifact of a particular aggregation formula: it can arise when predictive and contextual information are allowed to compensate for one another. The specific threshold Φ_A depends on the operator and on the two alternatives being compared. The theorem does not imply that every compensatory operator, or every pair of alternatives, admits a reversal within the unit interval.

This distinction is particularly relevant in CADEMAs-ML because the two aggregated components do not represent equivalent evaluation criteria. The predictive score R_i summarizes evidence generated by machine-learning models, whereas the contextual score Q_i represents compliance with organizational, regulatory, or domain-specific requirements. Therefore, the choice of aggregation operator implicitly defines the degree to which contextual requirements can modify or override data-driven recommendations.

4.5. Sensitivity to Predictive and Contextual Perturbations

In many situations, both predictive scores and contextual assessments could be subject to uncertainty. Predictive models may produce unstable estimates due to limited data, model uncertainty, or distribution changes, whereas contextual assessments may vary because of incomplete knowledge, expert disagreement, or evolving organizational requirements.

Therefore, an additional relevant property of an aggregation mechanism is its sensitivity to perturbations in its input dimensions. In the context of CADEMAs-ML, this property determines how strongly changes in predictive evidence or contextual requirements affect the final prioritization score.

Let $A : [0, 1]^2 \rightarrow [0, 1]$ be an aggregation operator. For perturbations ϵ_R and ϵ_Q applied to the predictive and contextual assessments, respectively, define the resulting variation in the prioritization score as

$$\Delta_A(\epsilon_R, \epsilon_Q) = |A(R + \epsilon_R, Q + \epsilon_Q) - A(R, Q)|. \quad (18)$$

Argument-specific perturbations are obtained by setting one of the two perturbations to zero:

$$\Delta_R(\epsilon) = |A(R + \epsilon, Q) - A(R, Q)|, \quad \Delta_Q(\epsilon) = |A(R, Q + \epsilon) - A(R, Q)|. \quad (19)$$

These quantities measure how changes in each information source propagate to the final decision score. Operators with bounded sensitivity provide a direct control on score-level perturbations, although such bounds do not by themselves guarantee the stability of the resulting ranking or Top-K selection.

Theorem 3 (Lipschitz sensitivity of aggregation operators). *Let $A : [0, 1]^2 \rightarrow [0, 1]$ be an aggregation operator continuously differentiable on an open neighborhood containing the line segment between (R, Q) and $(R + \epsilon_R, Q + \epsilon_Q)$. Assume that both endpoints, and, hence, the whole segment by convexity, lie in $[0, 1]^2$. If there exist finite constants L_R and L_Q such that*

$$\left| \frac{\partial A}{\partial R} \right| \leq L_R \quad \text{and} \quad \left| \frac{\partial A}{\partial Q} \right| \leq L_Q, \quad (20)$$

throughout that open neighborhood, then

$$|A(R + \epsilon_R, Q + \epsilon_Q) - A(R, Q)| \leq L_R |\epsilon_R| + L_Q |\epsilon_Q|. \quad (21)$$

Proof. By the Mean Value Theorem, there exists a point (ξ_R, ξ_Q) on the line segment between (R, Q) and $(R + \epsilon_R, Q + \epsilon_Q)$ such that

$$A(R + \epsilon_R, Q + \epsilon_Q) - A(R, Q) = \frac{\partial A}{\partial R}(\xi_R, \xi_Q) \epsilon_R + \frac{\partial A}{\partial Q}(\xi_R, \xi_Q) \epsilon_Q. \quad (22)$$

Taking absolute values and applying the derivative bounds (20) yields

$$|A(R + \epsilon_R, Q + \epsilon_Q) - A(R, Q)| \leq L_R |\epsilon_R| + L_Q |\epsilon_Q|. \quad (23)$$

□

The theorem establishes a local robustness bound for simultaneous perturbations of the two inputs. The constants L_R and L_Q quantify the maximum amplification of perturbations in the corresponding dimensions over the region considered. They do not establish a global ordering of aggregation operators or guarantee the stability of a population-level Top-K decision. Those questions are examined empirically in Section 5.

For the linear aggregation operator, $A_L(R, Q) = \lambda R + (1 - \lambda)Q$, the sensitivity coefficients are constant:

$$L_R = \lambda, \quad L_Q = 1 - \lambda. \quad (24)$$

Therefore, the influence of predictive and contextual perturbations is uniform throughout the decision space. A perturbation in either input has a marginal effect determined only by its corresponding weight. This property is consistent with the fully compensatory behavior of the linear operator, since the effect of one dimension does not depend on the current value of the other dimension.

For the weighted geometric operator, $A_G(R, Q) = R^\lambda Q^{1-\lambda}$, the local sensitivities are

$$\frac{\partial A_G}{\partial R} = \lambda R^{\lambda-1} Q^{1-\lambda}, \quad \frac{\partial A_G}{\partial Q} = (1-\lambda) R^\lambda Q^{-\lambda}. \quad (25)$$

Unlike the linear operator, these sensitivities depend on the current decision state. In particular, for $R > 0$, $\partial A_G / \partial Q$ becomes unbounded as $Q \rightarrow 0^+$. Similarly, for $Q > 0$ and $\lambda < 1$, $\partial A_G / \partial R$ becomes unbounded as $R \rightarrow 0^+$. Thus, finite Lipschitz constants for A_G are available only on regions bounded away from both coordinate axes. The operator can, therefore, exhibit strong local sensitivity when either input approaches zero. This property is relevant to uncertainty propagation near contextual vetoes, although it does not by itself determine the probability of a policy violation under noisy observations.

For the minimum operator, $A_M(R, Q) = \min(R, Q)$, the final score is determined by the limiting component. Consequently, the operator exhibits zero compensation: perturbations of the non-limiting input have no effect as long as the perturbation does not change which input is active. This provides strong preservation of the limiting constraint, while also making the response to uncertainty dependent on the current position relative to the boundary $R = Q$. Because A_M is not differentiable on $R = Q$, it is not covered by Theorem 3 at that boundary. The following corollary characterizes its piecewise-stable regimes.

Corollary 3 (Minimum operator stability under inactive-component perturbations). *Consider a decision case evaluated under the minimum operator $A_M(R, Q) = \min(R, Q)$.*

If $R < Q$ and a contextual perturbation ϵ_Q satisfies $R \leq Q + \epsilon_Q$, then

$$|A_M(R, Q + \epsilon_Q) - A_M(R, Q)| = 0. \quad (26)$$

Similarly, if $Q < R$ and a predictive perturbation ϵ_R satisfies $Q \leq R + \epsilon_R$, then

$$|A_M(R + \epsilon_R, Q) - A_M(R, Q)| = 0. \quad (27)$$

Proof. If $R < Q$, then

$$A_M(R, Q) = R.$$

Under the condition $R \leq Q + \epsilon_Q$, the predictive score remains the limiting component after the contextual perturbation. Hence,

$$A_M(R, Q + \epsilon_Q) = \min(R, Q + \epsilon_Q) = R,$$

and, therefore,

$$|A_M(R, Q + \epsilon_Q) - A_M(R, Q)| = 0.$$

The second statement follows analogously. If $Q < R$, then

$$A_M(R, Q) = Q.$$

Under the condition $Q \leq R + \epsilon_R$, the contextual score remains the limiting component after the predictive perturbation. Consequently,

$$A_M(R + \epsilon_R, Q) = \min(R + \epsilon_R, Q) = Q,$$

which gives

$$|A_M(R + \epsilon_R, Q) - A_M(R, Q)| = 0.$$

□

The corollary illustrates the asymmetric robustness of the non-compensatory minimum operator. Perturbations of the non-limiting component are completely absorbed while the limiting component remains unchanged. Once a perturbation changes the active constraint, however, the resulting score variation can be transmitted directly to the aggregate. This transmission is never amplified, since

$$|\min(R + \epsilon_R, Q + \epsilon_Q) - \min(R, Q)| \leq \max\{|\epsilon_R|, |\epsilon_Q|, |\epsilon_R - \epsilon_Q|\}.$$

In particular, the minimum operator is globally 1-Lipschitz with respect to each argument separately. Its sensitivity is, therefore, zero when the perturbed argument remains inactive and, at most, the magnitude of the perturbation when that argument becomes active.

The contrast with A_G is relevant. The geometric operator is smooth inside the positive quadrant but its local sensitivity can grow without bound near either coordinate axis. The minimum operator is non-smooth at $R = Q$ but has a uniform global sensitivity bound. Smoothness and robustness are, therefore, distinct properties of an aggregation mechanism.

5. Simulation Results

The theoretical analysis establishes how the aggregation operators differ in terms of compensation, veto preservation, and sensitivity to perturbations. The simulation study complements these score-level results by examining their consequences when the aggregated scores are used to prioritize a population of competing alternatives. The experiments, therefore, focus on two aspects that cannot be determined from the operator definitions alone: how the resulting prioritization changes as the relative weight of predictive and contextual evidence varies, and how stable the induced rankings and Top- K selections remain under perturbations of the underlying scores.

The experimental design uses controlled Monte Carlo populations to isolate these effects while keeping the decision architecture fixed. The analysis is organized around three complementary aspects. First, the sensitivity of the aggregation semantics to λ is examined, with particular attention to the compensatory behavior of the linear and geometric operators. Second, the stability of the induced rankings and Top- K selections are evaluated under perturbations of the predictive score R . This analysis treats R as the uncertain input, reflecting the predictive nature of the ADM outputs in the CADEMAs-ML setting. Although contextual assessments Q could also be uncertain in other implementations, a separate Q -noise experiment is unnecessary here because the aggregation operators exhibit the corresponding argument-wise properties established theoretically. The conclusions obtained for perturbations of R , therefore, extend to perturbations of Q under the analogous conditions. Finally, the effect of the intervention capacity is examined by varying K , allowing policy compliance to be assessed across different Top- K thresholds and noise levels. Throughout the analysis, the empirical findings are interpreted in relation to the theoretical properties established in Section 4.

5.1. Experimental Design

The simulation study uses controlled Monte Carlo populations to isolate the effects of aggregation semantics while keeping the underlying decision architecture fixed. Each population contains $N = 1000$ alternatives, with each alternative i characterized by a

predictive score $R_i \in [0, 1]$ and a contextual assessment $Q_i \in [0, 1]$. The same population is evaluated under the linear, weighted geometric, and minimum aggregation operators,

$$P_i = A(R_i, Q_i; \lambda),$$

which determines the resulting prioritization and Top- K selection. Unless otherwise stated, $K = 100$ and 20% of the population is assigned a contextual veto ($Q_i = 0$). Vetoed alternatives are generated with comparatively high predictive scores, creating explicit conflicts between predictive evidence and contextual admissibility. The remaining scores are sampled from bounded distributions designed to provide both aligned and conflicting combinations of R_i and Q_i , including weak-context cases with $0 < Q_i \leq 0.25$.

Unless otherwise stated, the linear and weighted geometric operators are evaluated at $\lambda \in \{0.10, 0.50, 0.90\}$ representing low, balanced, and high relative weight on predictive evidence, respectively, while avoiding an unnecessarily large number of pairwise comparisons. The minimum operator A_M is independent of λ and, therefore, provides a natural reference. The seven representative aggregation configurations are denoted as $A_L^{0.10}$, $A_L^{0.50}$, $A_L^{0.90}$, $A_G^{0.10}$, $A_G^{0.50}$, $A_G^{0.90}$, and A_M .

All experiments use 1000 Monte Carlo replications with registered seeds 42–1041.

In the first experiment, the evaluation considers two complementary aspects of the resulting prioritization: contextual policy compliance and decision stability. Let $\mathcal{T}_K$ denote the selected Top- K set and let

$$\mathcal{V} = \{i : Q_i = 0\}$$

denote the set of vetoed alternatives. Policy compliance is measured by the policy violation rate

$$V = \begin{cases} \frac{|\mathcal{T}_K \cap \mathcal{V}|}{K}, & K > 0, \\ 0, & K = 0, \end{cases}$$

which represents the proportion of selected alternatives that violate a contextual veto. The metric counts only strict vetoes ($Q_i = 0$) and does not treat weak non-veto cases ($0 < Q_i \leq 0.25$) as violations. Thus, $V = 0$ corresponds to complete preservation of the contextual vetoes, whereas larger values indicate greater policy non-compliance. In all reported experiments, K is fixed and positive, so V is evaluated through the $K > 0$ branch and is well defined for every experimental condition.

For the analysis about robustness to uncertainty in the predictive pathway, the predictive scores are perturbed according to

$$R'_i = \min\{1, \max\{0, R_i + \epsilon_i\}\}, \quad \epsilon_i \sim \mathcal{N}(0, \sigma_R^2), \quad (28)$$

and by comparing the resulting prioritization with the corresponding noiseless reference. This design models symmetric predictive uncertainty: perturbations can both inflate and deflate the observed scores after truncation to $[0, 1]$, rather than representing a one-directional overconfidence shift. The uncertainty analysis focuses on R because it represents the output of an ADM or predictive model and is, therefore, naturally subject to estimation error. No separate Q -noise experiment is required because the corresponding score-level sensitivity properties extend to Q by symmetry of the aggregation operators. Truncation to $[0, 1]$ is retained because both scores are bounded to $[0, 1]$.

Decision stability is assessed from two complementary perspectives. For a reference aggregated-score vector $P^{(1)} = (P_1^{(1)}, \dots, P_N^{(1)})$ and a perturbed or alternative vector $P^{(2)} = (P_1^{(2)}, \dots, P_N^{(2)})$, Kendall's τ is computed on the union of the two Top- K index sets, thereby measuring agreement among the alternatives that enter either prioritized tier (with

the tie correction inherent to the statistic). In contrast, the Jaccard similarity measures agreement between the corresponding Top- K sets,

$$J = \frac{|\mathcal{T}_K^{(1)} \cap \mathcal{T}_K^{(2)}|}{|\mathcal{T}_K^{(1)} \cup \mathcal{T}_K^{(2)}|}.$$

Kendall's τ , therefore, captures ordinal rearrangements among alternatives that compete for Top- K membership, whereas J focuses directly on the stability of the selected intervention sets. For Top- K selection, ties in P are resolved using the predictive score available in the corresponding condition as a secondary criterion and the alternative identifier as a final deterministic criterion, ensuring reproducible selections.

Monte Carlo means are reported together with 95% percentile intervals, which describe the dispersion of outcomes across independently generated populations rather than the sampling uncertainty of the estimated mean. The simulation study is, therefore, intended to characterize the population-level consequences of the analytically established aggregation properties, rather than to reproduce their proofs.

5.2. Policy Compliance Under Contextual Vetoes

The first experiment examines the population-level consequences of the contextual veto property established in Section 4.3. It addresses RQ2 by asking when the aggregation semantics preserve a contextual exclusion and when predictive evidence can compensate for it. The focus is, therefore, not on overall predictive performance, but on how the aggregation rule translates the two information sources into a prioritization.

Figure 1 gives the main population-level result. For low values of λ , all three operators produce zero observed policy violations. As predictive evidence receives more weight, however, the linear operator departs from this behavior. Its violation rate becomes positive at approximately $\lambda = 0.7$ and increases thereafter, reaching $V = 0.64$ at $\lambda = 1.0$ (95% percentile interval $[0.55, 0.73]$). The weighted geometric and minimum operators remain at $V = 0$ throughout the sweep. The onset around $\lambda = 0.7$ is specific to the simulated population and should, therefore, not be interpreted as a universal threshold for the linear operator.

The remaining panels of Figure 1 help trace this population-level pattern back to the individual scores. Figure 1 shows the first Monte Carlo population in the (R, Q) plane. The lines at $R = 0.5$ and $Q = 0.5$ separate broadly aligned cases, where predictive and contextual evidence point in the same direction, from conflicting cases, where one source is relatively strong while the other is weak. The vetoed alternatives form the horizontal $Q_i = 0$ band in the high- R region. This makes the intended stress condition visible directly because the cases that should be excluded are also among those that prediction would otherwise favor.

Figure 1 follow these same alternatives after aggregation. The dashed horizontal line marks the score of the K -th selected alternative, so points above it are competitive for Top- K selection. At $\lambda = 0.10$ and 0.50 , the vetoed cases remain outside this region for the linear operator, although their scores are already spread above zero. At $\lambda = 0.90$, their scores have moved sufficiently upward for some of them to reach the Top- K boundary. The weighted geometric and minimum operators show a different pattern: their vetoed cases remain at $P_i = 0$ independently of λ , while the non-vetoed population is redistributed above this boundary. The Figure, therefore, makes visible the mechanism behind the transition observed in Figure 1a: violations arise when a compensatory aggregation rule allows predictive evidence to lift a vetoed alternative into competition with admissible alternatives.

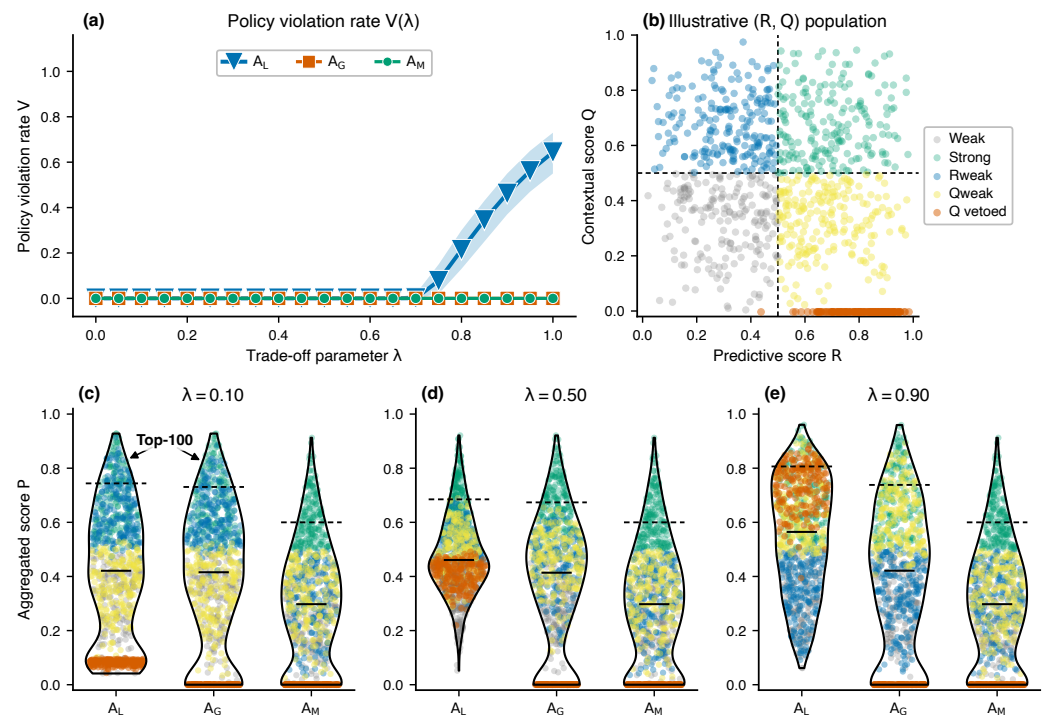


Figure 1. Policy compliance under contextual vetoes ($N = 1000$ and $K = 100$). (a) Monte Carlo mean policy violation rate over $\lambda \in [0, 1]$ across 1000 replications; shaded bands indicate 95% percentile intervals. (b) Illustrative population in the (R, Q) plane, including contextually vetoed cases ($Q_i = 0$) and non-veto cases with different predictive–contextual profiles. (c–e) Aggregated-score distributions for A_L , A_G , and A_M at $\lambda = 0.10$, 0.50 , and 0.90 , respectively, for the same population. The dashed line indicates the Top-100 cutoff and the overlaid points identify vetoed cases.

This distinction follows directly from the theoretical results. By Theorem 1, zero absorption implies that $Q_i = 0$ is propagated to a zero aggregated score. Corollary 2 then adds the population-level requirement needed to turn this score-level property into Top- K exclusion. Both conditions hold in the present design for A_G and A_M , explaining why their zero violation rate is deterministic rather than an empirical indication of superior performance. The linear operator does not satisfy zero absorption, so the theory provides no corresponding exclusion guarantee. Its behavior instead depends on the distribution of predictive and contextual scores, which is precisely what the dense sweep quantifies.

The consequences extend beyond explicit vetoes to the final prioritization. Figure 2 compares the rankings and Top- K sets produced by the seven representative configurations stated at the beginning. Kendall’s τ measures ordinal agreement restricted to the union of the two Top- K index sets, whereas Jaccard similarity measures the overlap between the selected Top- K sets. At balanced weighting, the compensatory operators with the same λ values remain relatively close: for example, $A_L(0.50)$ and $A_G(0.50)$ have mean $\tau > 0.88$ (Figure 2a). In contrast, configurations that combine different aggregation semantics and strongly different predictive weights can diverge substantially. The comparison between $A_L(0.90)$ and $A_G(0.10)$, for instance, gives mean $\tau < 0$ and a Jaccard overlap below 0.15. Thus, the choice of aggregation semantics can change not only whether a contextual veto is preserved, but also which alternatives are ultimately prioritized.

These agreement results should not be interpreted as a ranking of the operators themselves. They show, instead, that aggregation semantics define different mappings from the same (R_i, Q_i) evidence to the final decision order. The compliance results identify one particularly consequential manifestation of this difference: under a compensatory rule, increasing the predictive weight can make contextually excluded alternatives competitive;

under zero-absorbing rules, the theoretical exclusion property prevents this transition under the stated population conditions. Together, the two analyses connect the operator-level properties established theoretically with their population-level consequences for both policy compliance and the resulting Top- K prioritization.

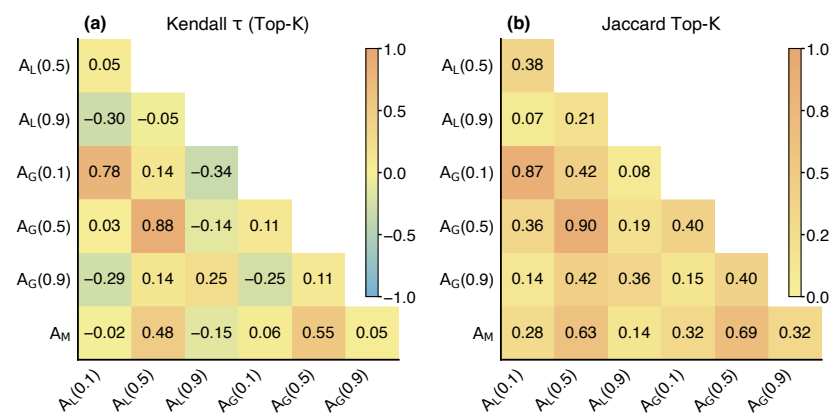


Figure 2. Pairwise agreement among the seven aggregation configurations considered in Experiment 01 ($N = 1000$, $K = 100$, and 1000 Monte Carlo replications). The configurations are A_L and A_G at $\lambda \in \{0.10, 0.50, 0.90\}$ and A_M . Each cell reports the Monte Carlo mean of the corresponding pairwise statistic across replications. (a) Kendall's τ on the union of the two Top- K index sets. (b) Jaccard similarity between the corresponding Top- K sets. Higher values indicate greater agreement; the diagonal is omitted because self-comparisons are identically equal to one.

5.3. Sensitivity Analysis

The second experiment examines how uncertainty in the predictive pathway propagates to the prioritization produced by CADEMAs-ML, addressing RQ3 and RQ4.

For each of the 1000 Monte Carlo populations, the predictive scores are perturbed according to Equation (28), with $\sigma_R \in \{0.05, 0.10, 0.20\}$. The clean case $\sigma_R = 0$ is retained as a baseline. The contextual scores remain unchanged, including the 20% of cases with $Q_i = 0$. The seven rankings derived from the seven aggregation configurations are evaluated.

For each configuration, the policy violation rate V is recorded, and the noisy ranking is compared with the corresponding clean ranking from the same population using Kendall's τ on the union of the two Top- K index sets and Top- K Jaccard similarity. The seven configurations are also compared under the same noisy realizations to distinguish deviations from their respective clean rankings from changes in agreement between aggregation semantics.

Figure 3 first illustrates the effect of predictive perturbations on the underlying population. Figure 3a–c show one representative Monte Carlo population at increasing noise levels. Because Q remains fixed, the perturbation acts horizontally in the (R, Q) plane. Cases subject to the contextual veto, therefore, remain on the $Q = 0$ boundary, whereas non-vetoed cases become increasingly dispersed along the predictive dimension. At the largest noise level, truncation to $[0, 1]$ also produces visible accumulation at the boundaries of the admissible interval. Thus, the perturbation is not an unconstrained Gaussian displacement, but a noisy predictive score bounded by $[0, 1]$ via Equation (28).

The population-level ranking consequences are summarized by policy compliance and ranking stability. The zero-absorbing operators A_G and A_M retain $V = 0$ across the tested noise levels, whereas perturbing R does not provide additional protection for the compensatory linear operator. In particular, the high-predictive configuration $A_L(0.90)$ continues to admit policy-violating cases. This follows directly from Theorem 1: perturbing R does not alter the property that A_G and A_M map a true contextual veto $Q_i = 0$ to zero. Since the population satisfies the supply condition of Corollary 2, this score-level property continues to exclude vetoed alternatives from the Top- K set.

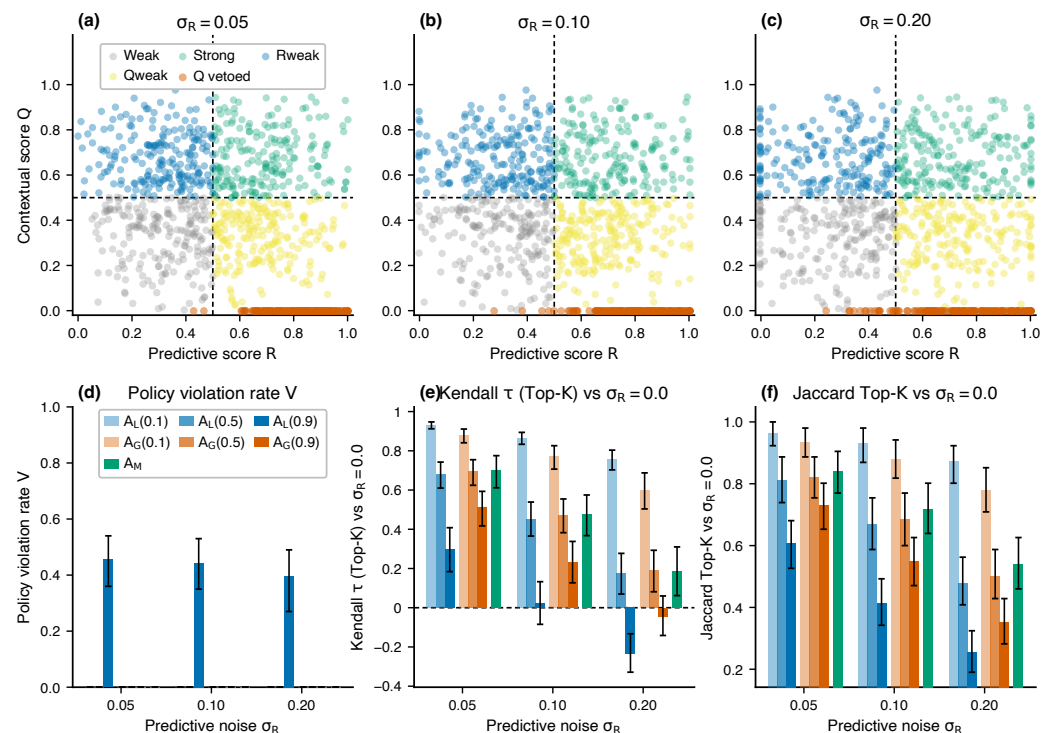


Figure 3. Sensitivity to predictive-score perturbations. Panels (a–c) illustrate one representative population at increasing noise levels, while panels (d–f) summarize the corresponding Monte Carlo effects on the seven ranking configurations (V , Kendall τ on the Top- K union, and Jaccard Top- K).

Predictive uncertainty also reduces agreement with the corresponding clean rankings. At $\sigma_R = 0.05$, most configurations exhibit a moderate degradation, which becomes substantially stronger as the noise increases. The effect is configuration-dependent, with settings that assign greater weight to R generally being more sensitive, particularly for the linear operator. At $\sigma_R = 0.20$, for example, the mean Kendall correlation for $A_L(0.90)$ falls to approximately -0.24 , while its Top- K Jaccard similarity is only about 0.25. Thus, large predictive perturbations can substantially rearrange the Top- K union and alter the selected intervention set. The stronger degradation in Jaccard than in Kendall's τ further indicates that the instability is particularly consequential around the Top- K boundary.

Although the minimum operator is 1-Lipschitz with respect to each argument, this global bound does not prevent ranking changes when the active component changes; the observed sensitivity reflects this boundary-crossing behavior rather than a violation of the Lipschitz property.

The agreement between aggregation configurations under shared noisy realizations is shown in Figure 4. Unlike the previous analysis, which compares each noisy ranking with its own clean counterpart, these heatmaps compare different aggregation semantics under the same perturbed predictive scores. Agreement generally decreases as σ_R increases, although the magnitude of the effect varies across configurations. In particular, settings with different λ values can diverge substantially, whereas configurations with similar integration semantics tend to remain more closely aligned. Predictive uncertainty, therefore, does not erase the effect of aggregation semantics; rather, it interacts with the way each operator transforms the perturbed evidence into prioritization scores.

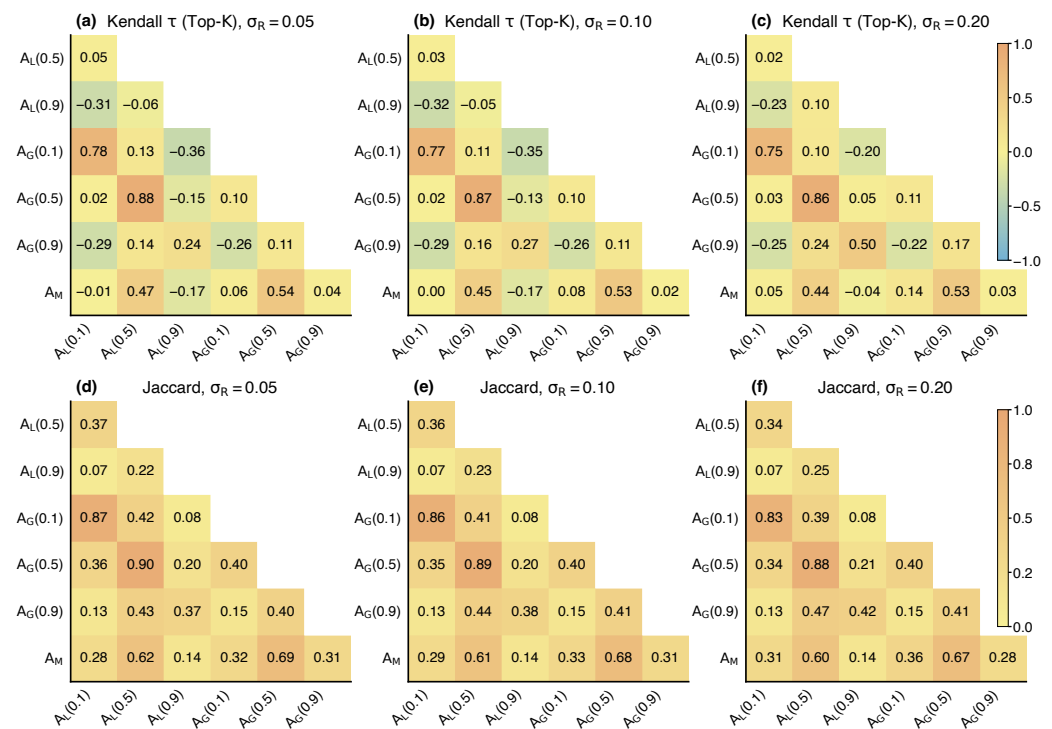


Figure 4. Pairwise agreement among the seven ranking configurations under shared predictive-score perturbations. The heatmaps report mean Kendall τ on the union of the two Top-K index sets and Top-K Jaccard similarity across Monte Carlo replications.

5.4. Intervention-Capacity Analysis

A complementary analysis examines how the intervention capacity K interacts with predictive uncertainty. Figure 5 reports the mean policy violation rate V as a function of K for the four predictive-noise levels $\sigma_R \in \{0, 0.05, 0.10, 0.20\}$ used before.

The shaded regions represent the corresponding 95% Monte Carlo intervals. For the zero-violation configurations, V remains at the zero baseline across the tested noise levels and values of K , reflecting the structural preservation of the contextual veto. The same near-zero behavior is observed for the remaining configurations except for the compensatory linear operator $A_L(0.90)$, which places the greatest emphasis on predictive evidence.

For $A_L(0.90)$, the effect of intervention capacity is particularly informative. Under the clean baseline, policy violations are already present at $K = 10$ and remain visible as K increases. As predictive noise increases, however, the onset of violations is progressively delayed: the corresponding curve shifts toward larger K values. At the maximum perturbation, $\sigma_R = 0.20$, the violation rate remains near zero for the smaller intervention capacities and begins to increase only beyond, approximately, $K = 50$. This behavior does not indicate the improved policy compliance of the aggregation operator. Instead, it reflects a change in the composition of the cases reaching the Top-K cutoff under noisy predictive scores.

This compositional effect is made explicit in Figure 5e, which decomposes the Top-K population for $A_L(0.90)$ into three mutually exclusive categories: *clipped non-vetoed* cases, *Q-vetoed* cases, and *other* cases. Here, a *clipped non-vetoed* case is a non-vetoed case ($Q_i > 0$) for which $R_i + \epsilon_i$ falls outside $[0, 1]$, so that R'_i is truncated to an admissible boundary (0 or 1); the *Q-vetoed* category comprises all cases with $Q_i = 0$, and the *other* category contains the remaining cases. Each tile represents the complete Top-K set for a given combination of K and σ_R , so the three proportions sum to 100%.

In the clean condition (i.e., $\sigma_R = 0.0$, Figure 5e), clipped non-vetoed cases are virtually absent, whereas *Q-vetoed* cases are already represented at small values of K and

become increasingly prevalent as the intervention capacity grows. Once predictive noise is introduced, clipped non-vetoed cases enter the prioritized set and progressively displace Q -vetoed and other cases. The effect is strongest at $\sigma_R = 0.20$: clipped non-vetoed cases account for approximately 90.5% of the Top-K set already at $K = 10$. Their proportion subsequently decreases as K increases, although it does not disappear completely. From approximately $K = 60$ onward, Q -vetoed cases begin to reappear and their representation increases with K , reaching about 39.7% at $K = 100$.

The composition analysis, therefore, explains the apparently delayed increase in V observed for $A_L(0.90)$ under strong predictive noise. Truncation to $[0, 1]$ concentrates perturbed predictive scores at the admissible boundary and changes which cases occupy the highest positions in the ranking. Policy-violating cases consequently do not enter the Top-K until the intervention capacity is sufficiently large for the Q -vetoed cases to re-emerge among the selected alternatives. The V -versus- K analysis, thus, complements the noise-based ranking measures without introducing a new source of uncertainty: it reveals how intervention capacity and score perturbation jointly affect the composition of the selected set.

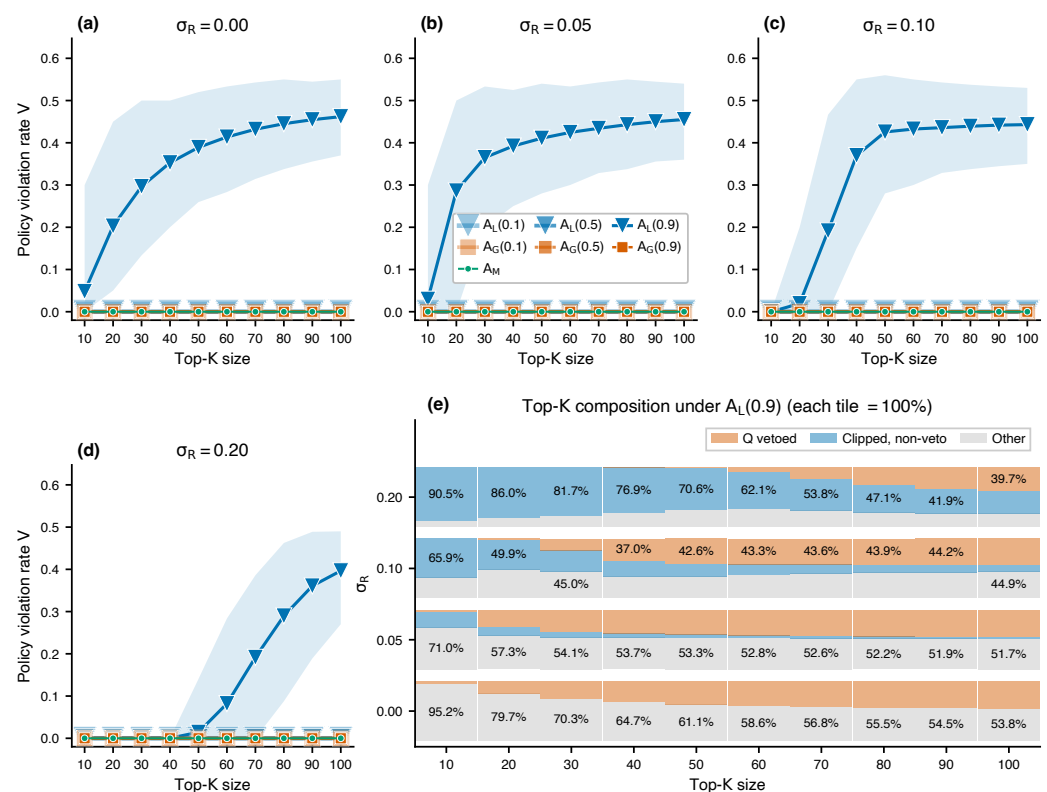


Figure 5. Sensitivity of policy compliance to intervention capacity and predictive uncertainty. Panels (a–d) show the mean policy violation rate V as a function of Top-K size for $\sigma_R \in \{0, 0.05, 0.10, 0.20\}$, with shaded 95% Monte Carlo intervals for the seven ranking configurations. Panel (e) shows the corresponding Top-K composition for $A_L(0.90)$ as a function of K and σ_R : each tile represents the percentage of *clipped non-vetoed*, *Q-vetoed*, and *Other* cases in the selected set.

Taken together, the sensitivity analyses reveal two complementary consequences of predictive uncertainty. First, noise can substantially reduce ranking stability, particularly for configurations that assign greater weight to predictive evidence. Second, its effect on policy compliance depends on both aggregation semantics and intervention capacity. Zero absorption preserves contextual vetoes independently of the predictive perturbation, whereas the compensatory linear operator can admit vetoed cases. For $A_L(0.90)$, the apparent delay in policy violations under strong noise is explained by the changing composition of the Top-K

set, driven in part by truncation at the boundaries of the predictive-score domain. Thus, uncertainty changes the stability and composition of the resulting prioritization, but does not alter the underlying aggregation semantics that determine whether contextual vetoes are structurally preserved.

6. Employee-Attrition Case Study

The previous experiments examined aggregation semantics under controlled population designs. This section complements that analysis with an employee-attrition case study based on the CADEMAs-ML workflow introduced in Godz et al. [7]. The objective is not to provide an independent validation of the underlying predictive or contextual models, but to examine how different aggregation semantics affect prioritization when the same employee-level assessments are used.

6.1. Case-Study Data and CADEMAs-ML Scores

The case study revisits the employee-attrition application originally introduced in CADEMAs-ML [7]. The cohort consists of $n = 100$ employees sampled from the *IBM HR Analytics Employee Attrition and Performance* dataset (Available at <https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset>, accessed on 10 July 2026). The present study reconstructs the original CADEMAs-ML workflow to obtain the employee-level predictive and contextual scores used in the subsequent aggregation analysis.

The predictive score R_i is obtained from four independent machine-learning ADM units representing different organizational perspectives: *Culture and Wellbeing*, *Finance*, *Operations and Development*, and *Human Resources*. Following the original application, each ADM unit is trained independently using its corresponding feature subset and H2O AutoML [25], with an 80/20 train-validation split and a one-hour time budget. The resulting models are exported as Model Object, Optimized (MOJO) artifacts and applied to the case-study cohort to obtain positive-class attrition probabilities $\hat{p}_{ik}$. The four predictions are then combined into the cooperative predictive score

$$R_i = \sum_{k=1}^4 w_k \hat{p}_{ik}, \quad (29)$$

where the weights are proportional to the validation area under the receiver operating characteristic curve (AUC) of the respective ADM unit,

$$w_k = \frac{\text{AUC}_k}{\sum_{\ell=1}^4 \text{AUC}_\ell}. \quad (30)$$

Thus, $R_i \in [0, 1]$ represents the integrated predictive assessment of employee-attrition risk while preserving the contributions of the four partially independent ADM units.

The contextual score Q_i is computed independently from the predictive layer through the *Digital/Technological Transformation* context defined in the original application. This context comprises seven fuzzy atomic rules concerning employee characteristics relevant to technological transformation. These rules are combined into four higher-level criteria: *upskilling capacity*, *career scalability*, *investment horizon*, and *role adaptability*. The resulting criteria are aggregated according to the context policy and combined conjunctively using the Gödel minimum. Consequently, failure to satisfy any of the four higher-level requirements yields $Q_i = 0$. The resulting $Q_i \in [0, 1]$, therefore, represents the degree of contextual alignment of employee i with the transformation scenario, independently of the predictive attrition assessment.

The resulting cohort contains 35 employees with $Q_i = 0$, corresponding to contextual vetoes, while the remaining employees have strictly positive contextual scores. The predictive scores span almost the entire admissible range, approximately $R_i \in [0, 0.99]$, whereas the non-vetoed contextual scores occupy a narrower range, approximately $Q_i \in [0.06, 0.73]$. This combination provides a heterogeneous decision population in which predictive risk and contextual alignment are only partially aligned. In particular, high predictive risk does not necessarily imply high contextual alignment, while contextual vetoes may occur at different levels of predictive risk. This makes the cohort suitable for examining whether the aggregation rule can alter prioritization when the underlying predictive and contextual assessments are held fixed.

6.2. Experimental Setup

The analysis considers a fixed intervention capacity of $K = 10$, corresponding to 10% of the $n = 100$ employee cohort. For each employee, the same predictive and contextual scores (R_i, Q_i) described above are combined using the seven integration configurations considered before: $A_L(0.1)$, $A_L(0.5)$, $A_L(0.9)$, $A_G(0.1)$, $A_G(0.5)$, $A_G(0.9)$, A_M . Thus, the predictive and contextual assessments, as well as all employee attributes, remain unchanged across configurations; only the integration semantics used to obtain P_i vary.

For each configuration, employees are ordered by decreasing prioritization score P_i , and the first K positions define the intervention set. Exact ties are resolved deterministically by first comparing the predictive score R_i and then using the original cohort order as a final tie-break. This convention ensures that every configuration produces a unique Top- K set and avoids introducing an additional policy-specific rule for tied prioritization scores.

The analysis compares both the resulting score distributions and the composition of the Top-10 sets. In particular, $A_L(0.5)$ is used as a reference configuration for the rank-trajectory analysis, allowing the effect of changing the aggregation semantics to be expressed as rank displacements for employees initially selected under the reference policy. Set-level differences are additionally quantified through the intersection and Jaccard similarity between the corresponding Top-10 sets.

For brevity, implementation-level details of the case-study construction are not reproduced here. The complete workflow, including the four predictive ADM units, fuzzy-rule-based contextual evaluation, computation of R_i and Q_i , all seven aggregation configurations, deterministic ranking procedure, and generation of the reported results, is implemented in the repository accompanying this paper. The implementation, therefore, provides a reproducible specification of the complete case-study pipeline while allowing the present section to focus on the consequences of the aggregation choices.

6.3. Results and Interpretation

Figure 6 provides an overview of the resulting decision population and prioritization scores. Specifically, Figure 6a shows that predictive risk and contextual alignment are only partially aligned in the cohort. In particular, the population contains a substantial group of contextually vetoed employees with $Q_i = 0$, while several employees with relatively high predictive risk exhibit only moderate contextual alignment. This separation between the two dimensions is important because it creates a setting in which the integration rule can materially affect the resulting priority rather than merely rescaling a common ordering.

Figure 6b shows the distributions of P_i produced by the seven configurations. The Top-10 cutoff varies considerably across operators, from approximately 0.33 for A_M to approximately 0.89 for $A_L(0.9)$. This variation reflects not only differences in score scale but also changes in the relative position of employees within the ranking. In particular, increasing λ in the linear operator progressively emphasizes predictive risk and shifts the prioritization

toward employees with high R_i . This effect is especially consequential for $A_L(0.9)$: one employee with $Q_i = 0$ enters the Top-10, despite being contextually vetoed. In contrast, the zero-absorbing operators A_G and A_M assign $P_i = 0$ whenever $Q_i = 0$ and, therefore, exclude all vetoed employees from the Top-10 in this cohort. The Figure, consequently, illustrates, at the application level, the structural distinction established in the theoretical and simulation analyses.

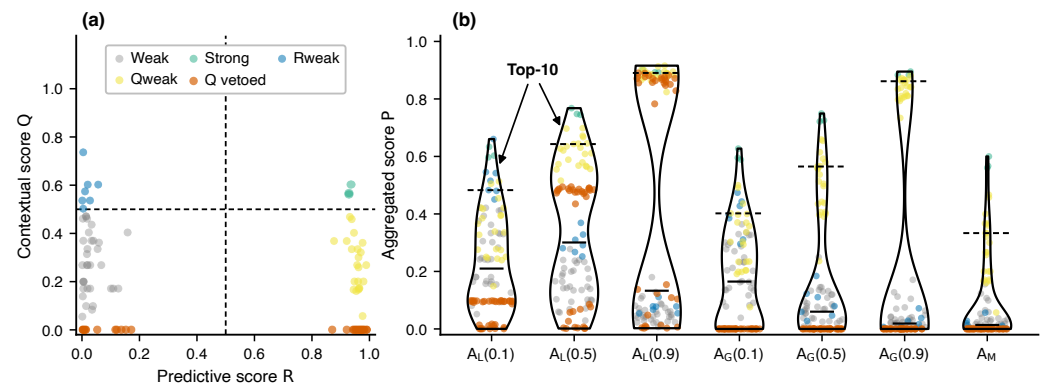


Figure 6. Employee-attrition case study ($n = 100$ and $K = 10$). (a) Cohort in the (R, Q) plane, with contextually vetoed cases ($Q_i = 0$) along the horizontal axis. (b) Aggregated-score distributions for the seven integration configurations. The dashed line on each violin is the Top- K cutoff ($K = 10$); overlaid points use the same stratum colours as panel (a).

The distributional differences translate directly into changes in the composition of the intervention set. To make these changes explicit, Figure 7 tracks the ten employees selected under $A_L(0.5)$ as the reference configuration. Under this policy, the selected employees are characterized by high predictive scores and non-zero, which are generally moderate-to-high contextual scores. Changing only the aggregation rule can, nevertheless, move several of these employees out of the intervention capacity. The Top-10 intersection with the reference set involves only five employees for $A_L(0.1)$ and six for $A_L(0.9)$, corresponding to Jaccard similarities of 0.33 and 0.43, respectively. Thus, both stronger context weighting and stronger predictive weighting can produce substantial replacement of intervention slots relative to the intermediate linear policy.

The effect is particularly pronounced for the high-predictive linear configuration. Several employees who occupy the middle or lower part of the reference Top-10 are strongly demoted under $A_L(0.9)$, while employees with very high R_i and comparatively weaker contextual alignment move upward. For example, Andrew Ross moves from rank 4 to rank 17 and Caroline Bennett from rank 5 to rank 14, whereas Noah Murphy moves from rank 7 to rank 2 and Christian Bennett from rank 9 to rank 3. The most pronounced displacement is observed for Madison Foster, whose rank changes from 10 to 37. These movements illustrate that the effect is not simply a change in the absolute scale of P_i : the aggregation semantics alter the relative ordering of employees and, consequently, which employees receive intervention.

The zero-absorbing configurations are comparatively stable around the intermediate weighting. The Top-10 Jaccard similarity reaches 0.82 for both $A_G(0.5)$ and A_M , with most employees from the $A_L(0.5)$ reference set retaining positions within or close to the intervention boundary. Nevertheless, even these operators can replace individual employees when their aggregation semantics differ sufficiently from the reference linear policy. Hence, zero absorption should not be interpreted as implying identical rankings; its structural guarantee concerns the treatment of contextual vetoes, not complete rank invariance.

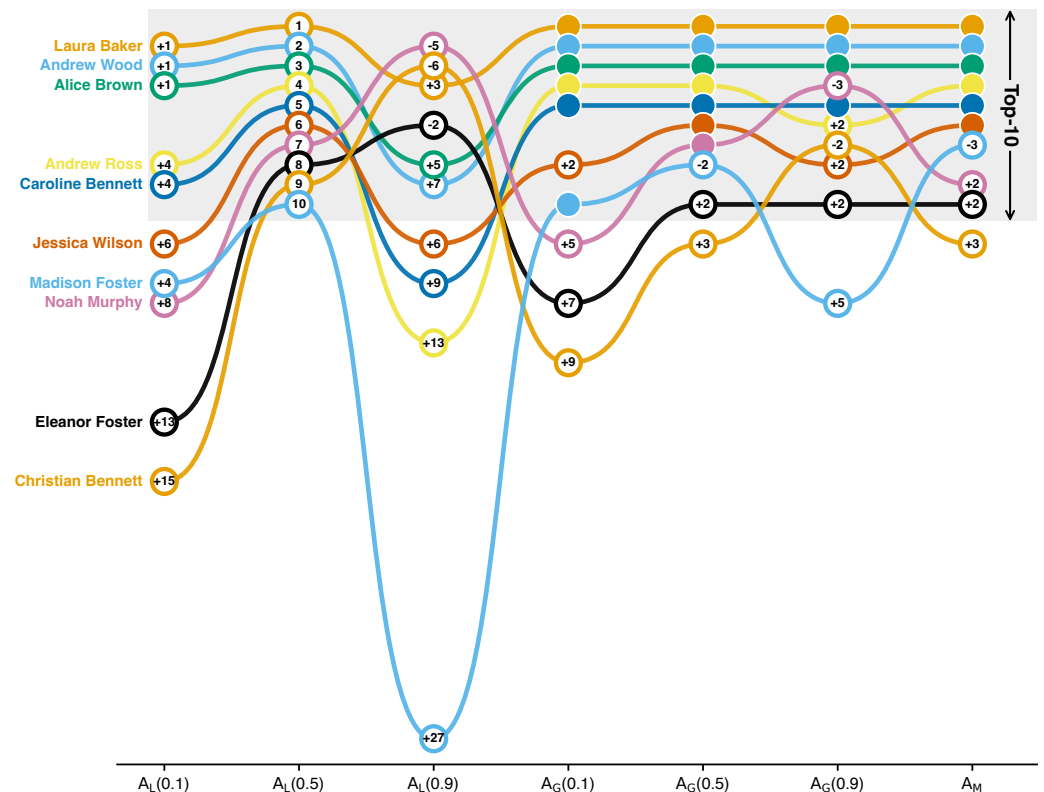


Figure 7. Rank trajectories of the Top-10 employees under $A_L(0.5)$ across the seven aggregation configurations ($n = 100$ and $K = 10$). Absolute ranks are shown at the reference column $A_L(0.5)$; elsewhere, markers report the signed change relative to that baseline ($\Delta = \text{rank} - \text{rank}_{A_L(0.5)}$), with solid series-coloured markers when the rank coincides with that at $A_L(0.5)$. The shaded band highlights the Top-10 region.

Figure 7 complements the distributional view by tracking the same Top-K intervention set under changes in aggregation semantics. The trajectories are those of the ten employees selected under $A_L(0.5)$; at the reference column the markers report absolute ranks, whereas the remaining columns report the signed rank displacement relative to that baseline (solid markers indicate an unchanged position). The shaded band marks the Top-10 region, so departures below that band correspond to employees that leave the intervention set under the corresponding operator.

7. Discussion and Conclusions

This study examined how aggregation semantics affect prioritization in Cooperative Automated Decision-Making Systems when predictive evidence and contextual requirements are integrated into a common decision score. Three aggregation regimes were considered: linear, weighted geometric, and minimum aggregation. Their structural properties were analyzed together with their effects on policy compliance, predictive uncertainty, ranking stability, and resource-constrained Top-K selection.

The results provide consistent answers to the four research questions. For RQ1, the analysis shows that aggregation semantics determine how predictive and contextual evidence interact. The three operators represent different compensation regimes and can, therefore, produce different prioritization outcomes from the same underlying evidence. For RQ2, zero absorption emerges as the key structural property for preserving contextual vetoes through the integration stage. Under the considered conditions, the weighted geometric and minimum operators preserve such vetoes, whereas the linear operator with

$\lambda \in (0, 1)$ allows predictive evidence to compensate for a null contextual assessment. The resulting policy behavior depends on both the aggregation semantics and the characteristics of the decision population.

The results for RQ3 show that predictive uncertainty can propagate from the predictive pathway to the final prioritization. Increasing perturbations of R reduce agreement with the corresponding clean rankings and can modify the composition of the Top- K intervention set. The theoretical sensitivity analysis provides a score-level explanation for differences in how perturbations are transmitted by the aggregation operators. For RQ4, the experiments indicate that uncertainty interacts with aggregation semantics rather than eliminating their differences. Ranking agreement between configurations can decrease as uncertainty increases, while the underlying policy behavior remains dependent on the selected aggregation rule. The observed policy-violation rate also depends on the intervention capacity and the resulting Top- K boundary.

The employee-attrition case study demonstrates that aggregation semantics can have a direct operational consequence in a realistic CADEMAs-ML workflow. The predictive and contextual assessments remain fixed throughout the comparison, yet changing only the integration operator alters the distribution of prioritization scores, the relative ranking of employees, and, ultimately, the composition of a resource-constrained intervention set. The case study, therefore, provides a concrete application-level illustration of the central result of the preceding analyses: the choice of aggregation policy is not a cosmetic modeling decision because it can determine which individuals receive scarce intervention resources.

The main implication for cooperative automated decision-making is that aggregation should be treated as an explicit policy component of the architecture. In CADEMAs-ML, predictive and contextual pathways can produce the same inputs while different aggregation rules generate different prioritization outcomes. This is particularly relevant when contextual assessments represent eligibility conditions, organizational requirements, or policy constraints.

This interpretation is consistent with the broader literature on context-aware and cooperative decision-support systems, where contextual knowledge is used to constrain or guide predictive decision processes. It also extends the perspective adopted in CADEMAs [6] and CADEMAs-ML [7] by showing that the integration stage itself can determine whether a contextual requirement remains effective after predictive and contextual information have been combined.

The results also distinguish policy compliance from ranking stability. An operator may preserve contextual vetoes while producing rankings that differ substantially from those obtained with another operator. Conversely, similar Top- K sets in a particular population do not imply equivalent policy semantics. This distinction is important when aggregation is used in resource-constrained decisions, since changes near the Top- K boundary can alter the alternatives that receive an intervention.

Several limitations delimit the scope of the conclusions. First, the analysis considers three aggregation regimes: linear, weighted geometric, and minimum aggregation. They represent distinct compensation semantics but do not cover the full range of fuzzy, nonlinear, or non-additive operators. The results should, therefore, not be generalized to aggregation families that were not evaluated.

Second, the Monte Carlo experiments use controlled populations designed to expose differences between predictive and contextual evidence. This design supports the analysis of specific structural properties, but it does not reproduce the full variability of deployed-decision environments. In particular, observed thresholds for policy violations depend on the population distribution and the selected Top- K capacity.

Third, the uncertainty experiments perturb the predictive score R . For the operators considered, the corresponding analysis for contextual perturbations follows from the symmetry of the aggregation functions and the complementary weighting parameter. This avoids a redundant experiment, but does not capture all forms of uncertainty in contextual reasoning, including uncertainty in rules, membership functions, or knowledge structures.

Several future work directions follow from these results. First, the analysis can be extended to richer aggregation families, including OWA operators, Choquet integrals, and other non-additive mechanisms. Learned aggregation is another relevant direction, particularly when aggregation parameters are estimated from historical decisions or stakeholder preferences. Such approaches would require an assessment of whether the resulting policies preserve the desired structural properties before deployment. Dynamic aggregation policies could also adapt the integration rule or its parameters to changes in context, uncertainty, resource availability, or decision objectives.

An additional direction concerns the interaction between aggregation levels within cooperative decision-making architectures. In CADEMAs-ML, aggregation may first occur within the individual decision pathways. For example, several ADMs may be combined to produce a predictive assessment, while multiple contextual rules or knowledge sources may be aggregated into a contextual assessment. The present study considers the subsequent integration of these two outputs into the final prioritization score. This creates a two-level aggregation structure whose properties have not been analyzed here. Future work could, therefore, investigate how the semantics adopted at the first aggregation level interact with those used at the second level. Such an analysis could reveal whether properties such as compensation, veto preservation, and sensitivity are preserved, amplified, or altered across aggregation levels.

Finally, larger and more heterogeneous real-world datasets would allow the interaction between aggregation semantics, uncertainty, and intervention constraints to be evaluated beyond the controlled and illustrative settings considered in this study.

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